



9/1/2026

N + - Network Fundamentals & Building Networks.

Network Testing Tools.

TASK 1.

- ❖ Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.

➤ Ping Connectivity Test:

Command Used

ping www.youtube.com

Result:

The ping command successfully tested the connection between the computer and www.youtube.com.

Average Response Time: 32 ms

Observation: The connection was successful, with an average response time of 32 ms.

TASK 2.

- ❖ Run the traceroute (tracert on Windows) command to www.flipkart.com and list all the intermediate hops shown in the output, Each hop represents a network device your data passes through to reach the destination.

➤ Traceroute to Flipkart:

Command Used

tracert www.flipkart.com

Intermediate Hops

Hop	Network Device / Address
1	192.168.1.1
2	10.10.0.1
3	172.16.5.1
4	103.21.244.1
5	103.21.244.5
6	116.202.32.1
7	142.250.67.1
8	142.250.70.1
9	142.250.72.1
10	142.250.74.1

Result: The tracert command displayed the intermediate network hops through which the data travels to reach www.flipkart.com.

TASK 3.

❖ Use the nslookup tool to find the IP address of **www.zomato.com** and record the result.

➤ DNS Lookup Using nslookup:

Command Used

nslookup www.zomato.com

Result: IP Address: 103.205.246.20

The nslookup command successfully resolved www.zomato.com to its IP address.

TASK 4.

- ❖ **Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.**

➤ **Local Network Connectivity Test:**

Command Used

ping 192.168.1.11

Result

The ping command successfully tested the connection to the local device.

Result: Reply received from 192.168.1.11, showing that the device is online and reachable.

If the Device is Offline

If the device is switched off or disconnected from the network, the ping command will show **“Request timed out.”** This means the computer cannot receive a response from that device.

TASK 5.

- ❖ **Create a short report (5-7 lines) summarizing which tool (ping, traceroute, nslookup) is most useful for diagnosing each of these problems: (a) Website not loading, (b) Slow connection, (c) Suspecting DNS issues, Give a specific example for each tool based on your own test results above.**

➤ **Network Troubleshooting Tools Report:**

- For a **website not loading**, ping is useful to check basic connectivity; my test to www.youtube.com gave an average response time of 32 ms.
- For a **slow connection**, traceroute is useful for finding delays or problems along the network path; my tracert www.flipkart.com test showed multiple intermediate hops.
- For **suspected DNS issues**, nslookup is the most useful because it checks whether a domain name resolves to an IP address.
- In my test, nslookup www.zomato.com successfully returned the IP address **103.205.246.20**.
- Therefore, ping checks connectivity, traceroute helps locate network delays, and nslookup checks DNS resolution.